

BIOLOGY — GRADE 11 | ECOLOGY

Quick Check

Lesson 5: Belt Transect Across a Changing Habitat: How Does Plant Population Change as Conditions Change?

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Ecology | Lesson 5

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Before collecting data, a learner predicts that a shaded area of the school compound will have fewer grass plants per square metre than the open field. Which reason best supports this prediction?

- A** Plants in the shade do not need any water
- B** Grass grows best when it gets no sunlight at all
- C** Shade gives less sunlight, so the grass can make less food by photosynthesis
- D** Shaded areas always have more trampling than open fields

In a belt transect study, a group places quadrats at regular intervals (every 2 m) between two parallel lines instead of choosing the greenest spots. Why is this done?

- A** It avoids bias, so the sample represents the whole transect and not only the best places
- B** It makes sure every quadrat contains the same number of plants
- C** It protects the plants from being trampled by the learners
- D** It makes the counting finish faster because there are fewer plants to count

Team A counted the plants in five 1 m x 1 m quadrats placed in the open field: 16, 20, 18, 22 and 14 plants. What is the mean abundance per quadrat in the open field?

A 22 plants

B 18 plants

C 90 plants

D 20 plants

A group uses a 50 cm x 50 cm quadrat (area 0.25 m²) in the garden zone. The mean count is 3 plants per quadrat. What is the estimated density in plants per square metre?

A 0.75 plants per m²

B 6 plants per m²

C 12 plants per m²

D 3 plants per m²

The school pathway gets plenty of sunlight, but the transect showed very few plants growing on it. What is the best explanation?

- A** Plants on pathways have no competition, so they cannot grow
- B** Too much sunlight kills all plants
- C** Constant trampling damages plants and compacts the soil, so few plants can grow
- D** Pathways have no soil at all

In the school garden, manure and water were added and weeds grew very thickly. A river near a farm receives fertiliser and sewage, and algae and aquatic weeds grow thickly there too. Which statement best links the two situations?

- A** Fish eat the algae, so the algae are the cause of the fish disappearing
- B** Extra nutrients boost the growth of plants and algae in both places, as the manure does in the garden
- C** Extra nutrients reduce the growth of algae in rivers but increase it in gardens
- D** Algae and weeds in the river are many, so the river must be healthier now

Which pair of factors that could affect fish in a river contains one abiotic factor and one biotic factor?

A Oxygen level in the water and competition for food among fish

B Algae and aquatic weeds

C Water temperature and nutrient level

D Sunlight and water temperature

In the shaded part of the school compound, few plants grew because there was little light. In ecology, what do we call a condition like this that restricts how large a population can grow?

A A quadrat

B A limiting factor

C A habitat

D A niche

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).